

MODULE IV

SITE PLANNING AND BUILDING RULES

INTRODUCTION

The location of the building (i.e., the site) has utmost importance in bringing a tranquil atmosphere in a dwelling. A properly located site with good foundation soil brings down the cost of construction. Natural ventilation and lighting, vicinity of other utility centres, transportation and communication facilities govern the selection of a site for residence. These factors, which are to be considered for selecting a site for a building is explained in the next section.

SELECTION OF SITES

The various factors that are to be considered for the selection of site for a residential building are

1. The site should be fairly level with good qualities of soil
2. The location should be calm but reasonably developed
3. It should be well connected by the roads and other modes of transport
4. It should have good communication facilities.
5. Electricity, water and sewer line should be available.
6. It should be away from hazardous industries.
7. Flood prone areas, water logged areas and reclaimed land should be avoided.
8. The site should have good natural ventilation and lighting.
9. The amenities like schools, recreation centres, shopping centres, hospitals should be near.
10. Site should have quick drainage properties.
11. It may have regular shape with sufficient frontage.
12. The area must be sufficient for present and future developments.
13. The place prone to air and water pollution should be avoided.
14. The proper ownership and other legal matters have to be checked before buying a site.
15. The type of land use recommended at proposed site should be checked with local authorities

SITE PLAN

The detailed plan of the plot with the necessary surrounding data is called site plan. It may contain the following details.

1. Boundaries of the plot with revenue survey particulars
2. Position of site in relation to the neighbouring streets with its main access.
3. Details of street with name, width.
4. All existing structures in the plot.
5. All existing streets and footpaths within the plot.
6. Layout of streets adjoining, terminating at the site, existing, proposed to be widened or newly aligned.
7. Proposed subdivision of plot, if any.
8. The access to each plot subdivisions
9. Details of undeveloped, not proposed to be developed and to be reclaimed area within the plot.
10. Area and location of any paddy field and other agricultural land that are proposed to be converted.
11. North direction and predominant wind direction.
12. Layout of proposed building (Details not required).

13. Topographic contours.

All these details are to be plotted in the site plan to a scale of not less than 1:400 in paper of size not less than 24cm X 34cm .

This site plan is to be submitted along with key map and service plan to the municipal or the town planning authorities to obtain the sanction for the construction. While preparing the site plan, the prevailing building rules and by laws are to be taken into consideration. A typical site plan is shown in the figure.

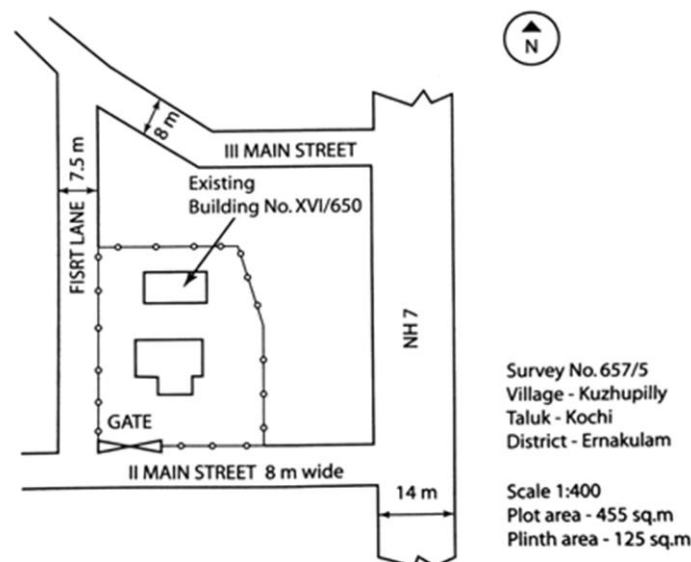


Fig.4.1:Site Plan

KERALA MUNICIPAL BUILDING RULES

The important regulations of Kerala Municipality Building Rules, 1999 regarding site and building requirements, exterior and interior open air spaces, coverage and floor area ratio, provisions of size, height and ventilation of rooms are given below. A number of amendments and modifications have made on Kerala Municipality Building Rules, 1999 within this period. All the amendments up to March 2015 are incorporated in this notes.

a) General requirement regarding plot

- i. Building should not be constructed on the land where refuse, excreta or other offensive matter is deposited.
- ii. Plots with big pits, quarries or similar excavations should be avoided unless it is levelled and made suitable.
- iii. Floodable areas and plots with a slope of more than 45 degrees should not be used for building construction.
- iv. Construction activities in the areas marked as coastal regulation zone under Environmental Protection Act is restricted.
- v. The minimum clearance from overhead electric lines for a building is given in table 4.1.

Table 4.1. Clearance from overhead electric lines

Sl.no	Type of line	Vertical clearance	Horizontal clearance
1	Low and medium voltage lines	2.4m	1.2m
2	Up to 33KV lines	3.7m	1.8m
3	More than 33KV	3.7m plus 0.3m for every additional 33KV	1.8m plus 0.3 m for every additional 33 KV

b) Exterior and interior open air spaces

Exterior open spaces are necessary for proper lighting and ventilation requirements.

Exterior open spaces (set backs)

- i. Every building up to 10m in height shall have a minimum front yard of 3m depth.
- ii. Every building up to 10m in height shall have a minimum rear yard of 2m depth.
- iii. Every building up to 10m in height shall have an open air space of not less than 1.2m width on one side and not less than 1.0m on other sides other than front and rear yards.
- iv. For buildings above 10m in height, 0.5m per increase of every 3m height is to be provided apart from minimum front, rear and side open spaces.

Table 4.2 :Exterior and Interior open space

Front open space	3.0m +0.5m per every additional height of 3m above 10m
Rear open space	2.0m+0.5m per every additional height of 3m above 10m
Side open space	1.2m +0.5m per every additional height of 3m above 10m

- v. For buildings above 4 storeys from ground level, a minimum open space of 5m is to be provided on any one side contiguous to front side.
- vi. The minimum distance from centre line of street to building should be 4.5m.
- vii. The minimum plot size for residential buildings should be 6m in width and 12m in depth.

Interior open spaces:

- i. Any habitable room not abutting on front, rear or side open spaces, shall abut an interior open space with a minimum width of 2.4m.
- ii. If the total height of the residential building is less than 7m, then the interior open space shall have 1.5m width.

c) Coverage and floor area ratio

DEFINITIONS

1. **Plot area:** Area enclosed between the boundaries of the plot.
2. **Plinth area:** It is the area of the building at plinth level excluding the open porch, uncovered staircase.
3. **Floor area:** It is the built up area of the building at any floor level.
4. **Carpet area:** It is the usable floor area excluding staircases, lift wells, ducts, toilets, electrical, air condition plant rooms etc.
5. **Coverage:** Coverage means the maximum area on any floor of the building excluding cantilevered open balconies, garden, compound wall, gate, uncovered staircase etc.

The maximum percentage of coverage (i.e. coverage/plot area as percentage) shall not exceed the limits shown in the table 4.3

Table .4.3 Coverage and Floor area ratio

Sl no	Building use	Maximum coverage in %	Maximum permissible F.A.R
1	Residential A1	65	3
2	Special Residential A2	65	2.5
3	Educational B	35	2.5
4	Medical/Hospital C	40	2
5	Assembly D	40	1.5
6	Office/Business E	40	2
7	Mercantile/ Commercial F	65	2.5
8	Industrial G1	40	1.5
9	Small Industrial G2	60	2.5
10	Storage H	60	2.5
11	Hazardous I(1)	30	1
12	Hazardous I(2)	25	0.7

6. **Floor area ratio (F.A.R):** The total floor area means total built up area in all floors excluding parking area. The ratio of total floor area on all floors to plot area is called Floor area Ratio.(F.A.R)

$$\text{Floor area ratio (F.A.R)} = \frac{\text{Total floor area on all floors}}{\text{Plot area}}$$

d) Provisions of size, height and ventilation of rooms

- a) The maximum height of the building is limited to a value equal to two times the width of the abutting road plus two times the front yard. This may be increased at a rate of 3 m for every additional 1m setback from building line.

- b) The minimum headroom in any human habitable room shall be 3.0 m. *For air-conditioned rooms it shall be 2.4m.
- c) The area of the habitable room shall not be less than 9.5m² with a minimum width of 2.4 m.
- d) Every habitable room shall be furnished with sufficient number of openings such as windows and ventilators to facilitate the direct admission of light and air.
- e) No portion of the room shall be assumed to be lighted if it is more than 7.5 m away from the opening.
- f) Every kitchen shall be ventilated according to the standards stipulated for habitable rooms and shall have suitable flue for escape of smoke and heated air.
- g) The area of the bathroom shall not be less than 1.50 sq m. With either side not less than 1.1 m, area of the latrine shall not be less than 1.10 sq m with one side not less than 1.0 m.
- h) Every bathroom or latrine shall have at least one of its walls abutting on an open air space or open verandah or an airshaft if not centrally air-conditioned; the size of air shaft shall not be less than that specified in table.4.4

Table:4.4 Size of air shaft

	No: of storeys	Area of the airshaft in sq.m	Minimum width of airshaft in m
1	Upto 3	1.08	0.9
2	Upto 6	2.4	1.2
3	Upto 10	3.0	1.5
4	Above 10	5.0	2.0

BASIC CONCEPTS OF INTELLIGENT BUILDINGS AND GREEN BUILDINGS

INTRODUCTION

In the modern world, the most emergent thoughts are about intelligent and green things. With the advent of artificial intelligence, smart programs are integrated to control of all most appliances, vehicles, machineries, other utilities. Automation is the key to artificial intelligence. Companies like Rockwell have automated vehicle production to deliver customer specific end products logistically sourcing parts and delivering finished products to a pre-determined programme.

In textiles jeans and shirts are being digitally customized to individual measurement-combining the advantages of mass production with the savoir of tailor-made perfection. It was only in the building industry that its arrival has been late and lethargic. However there are good efforts in this now; the following sections give an introduction to the intelligent building systems.

INTELLIGENT BUILDINGS

Though much has been discussed and practised in the field of intelligent building systems, a comprehensive definition of an intelligent building is yet to be established. However experts in this domain tried it. Various definition for the intelligent buildings are given below

“An intelligent building is in essence one that integrates various systems (such as lighting, heating, air conditioning, voice and data communication and other building functions) to effectively manage resources in a coordinated mode to maximize occupant performance, savings and flexibility minimising investment and operating cost”,

“Intelligent building combines innovations, technological or not, with skilful management to maximize returns on investments.”

“An intelligent building is one in which the building fabric, space, services and information systems can respond in an efficient manner to the initial and changing demands of the owner, the occupier and be in harmony with the environment”. Intelligent buildings are designed to increase efficiency of lighting, monitoring, safety and security, emergency systems, HVAC, door and window operations, guest registration and many more facets.

An intelligent building system concept considers the true cost of building is not its cost of construction; it must include the operating and maintenance costs over its life span. Intelligent buildings yield cost reductions over all these areas by optimizing energy use through automated control, communication and management systems. In a new intelligent building installation the following features are expected.

1. High-speed fibre optic communication network trunk for data, video and BAS (Building Administrative System)
2. Flexible HVAC system with modular distribution and 100% outdoor air capability to take advantage of free cooling as well as to allow flushing of the building to dilute volatile off-gassing contaminants.

3. Individually controlled HVAC terminal units allowing occupant control flexibility through Intelligent Terminals Controllers at each workstation.
4. Advanced integrated Energy Management and Control System (EMCS) utilizing direct digital control technology for HVAC, Lighting, Fire Alarm and other building support systems.
5. Dedicated circuit power distribution network complete with Uninterrupted Power Supply (UPS) Units.
6. Maximum transparency and communication capabilities between sub systems.
7. Electrical design features tailored to Intelligent Building.
8. Generous standby power generation.
9. High efficiency filtration, energy recuperation and / or thermal storage features to improve indoor air quality and energy consumption performance.
10. Networked multi-user access incorporating structured password protection.

Intelligent Building model

The Intelligent Building model structure has been broadly subdivided into seven systems which may be interfaced to varying degrees. These are:

1. Heating, Ventilating and Air Conditioning (HVAC) system
2. Electrical power distribution system.
3. Communications System
4. Lighting System
5. Vertical Transport System
6. Security System
7. Life Safety System

1. Heating, Ventilating and Air Conditioning (HVAC) system.

In Intelligent Buildings Systems the HVAC system must be selected in such a way that it satisfies ventilation standards and occupant comfort control while optimizing, flexibility, energy efficiency, maintenance costs. An owning and operating cost analysis coupled to an energy simulation of each viable option is mandatory to determine the optimal HVAC solutions.

The Personal Environment Controllers (PEC) are developed to be used in intelligent building to provide occupants with full control over their particular office works station environment. PEC units allow the occupant to control temperature, air flow volume and direction as well as lighting through their local PC or through hand held portable remote control units. In PEC concept, standard fixed position diffusers are replaced by one or more air jets strategically located to project cold air streams downward and away from the occupants. The occupant has control over the air volume and its direction.

2. Electrical power distribution System

The power distribution system deals with major electrical components and electrical energy monitoring. Major elements under monitoring and control include the emergency power generator, Uninterrupted Power Supply (UPS), the Emergency Lighting System, Individual tenant power metering units and other major electrical loads.

Emergency power system interfaces with other systems to annunciate normal power interruptions and initiate reactivation of emergency loads when emergency power becomes available.

It performs similar functions upon restoration of normal power including pre-programmed orderly reactivation of heavy electrical loads.

The Power Distribution system may also incorporate a load stabilization or load cycling program to minimize electrical demand charges.

Uninterrupted power source units are constantly monitored for critical elements such as normal/emergency power mode, trouble or alarm conditions etc.

3. Communication Systems

In the intelligent building, integrated communications is possible. The systems that may communicate include traditional LAN-based (Local Area Network) clients and servers, telephones and telephone switches, video conference devices and the full range of EMCS devices for HVAC, security, Lighting, and fire alarms. The intelligent building communication systems can also facilitate interchange with accounting or other networks to import/export pertinent data whether in-house or outsourced.

4. Lighting Systems

The lighting systems, in intelligent buildings must be integrated to the EMCS in order to interact with the other systems. When used with multilevel lighting control and perimeter zone daylight systems certain zones may also be tied to the load shedding program in the power distribution system. In intelligent buildings, the occupancy mode during extended hours may be controlled from local PC terminals or from strategically located override switches.

The extent of lighting control distribution, the use of motion detectors or infrared sensors and the use of lighting level sensors, in perimeter area, to allow use of natural lighting whenever possible must be incorporated. This lighting control should be extended to each room or workstation in order to take full advantage of the energy savings associated to non-occupancy mode or the limited and localized occupancy activations during extended hours for the HVAC and lighting systems. For example, Wipro lighting has been showcasing its energy saving lighting products like Light spot presence detection system and Managed Light system for corporate offices and Scene Select Systems for conference rooms.

5. Vertical Transport Systems

This system interacts with other systems such as the light safety or the security systems to define the number of elevator required, the mode of operation and accessible floor levels. To minimize energy consumption the number of activated elevators may also be reduced on the basis of time of day schedules, week or weekend days and statutory holidays.

6. Security Systems

Security system may include simple systems like automatic access monitoring, card access control, guard tour monitoring or motion detectors. More elaborate security systems involving intensive human intervention from guard stations and closed circuit TV and recording should be sourced from specialized security system manufacturers. The card access security program may be interfaced with lighting and HVAC subsystems to activate necessary lighting paths in specific room occupancy mode. Similarly the Life Safety Fire Alarm Program may be interfaced with security to

release specific locked doors under alarm conditions.

7. Life safety Systems

This subsystem deals with Fire Alarms the Emergency Lighting, Egress lighting system and Smoke Evacuation system. Interfacing of life safety system with other system is very important to eliminate nuisance alarms or to start the smoke evacuation system. Interfacing may also allow temporary transfer of spare emergency power not required at the time to other non-critical areas until such time as it is required.

THE CONCEPT OF GREEN BUILDINGS

Right from prehistoric man who constructed the huts for the first time to the great builder of the modern world has exploited the nature and its resources and destroyed the natural habits of other animals and birds. Now tremendous damages have been inflicted on our planet by the construction of various types of buildings using sand and water from rivers, stones from mountains, cement manufactured from the ingredients dug from the land. Also Carbon emission from the buildings and due to manufacture of construction materials warm up the air in space. This global warming leads to changes in climate, changes in rainfall pattern, rise in sea level, ozone depletion etc...On realizing the environmental responsibilities many efforts were put forth and one such intelligent initiative is the concept of "Green Building".

Green Buildings have a new approach to save water energy and material resources in the construction and maintenance of the buildings and can reduce or eliminate the adverse impact of building on the environment and occupants. *Green Buildings are eco-friendly structures.* This helps the earth and people to retain nature to a maximum extent the possible in three ways with reference to the location of the building as below.

1. Retain the external environment
2. Improve internal environment for the inhabitants
3. Reserve the environment at places far away from the building

1. Green Building Retain the Environment at the location of the Building

When a building is planned to occupy a vast area, the site selection should consider retention of local vegetation, wild life natural water courses etc.... Site with biodiversity should be avoided or the building should be planned to minimize site disturbance following aspects may be considered.

- i. **Land:** The landscaping and exterior design in a green building shall utilize more shaded area, the light trespass is eliminated and local species of plants are grown.
- ii. **Water:** The green building by its design and layout shall not disrupt the natural water flows, it should orient and stands just like a tree. Rain falling over whole area of the building shall be harvested in full either to replenish the ground water table or to be utilized in the services of the building. The toilets shall be fitted with low water consuming flushes the plumbing system should have separate lines for drinking and flushing. Grey water from kitchenette bath and laundry shall be treated and reused for irrigation or in cooling towers of air conditioning.
- iii. **Energy:** The solar energy on a green building is trapped to supplement the conventional energy. The natural light is allowed to enter in the intermediate floors to minimize the usage of electricity. Sunlight is restricted by the high grown trees outside the lower floors of the building. High efficiency light fixtures like LED lamp make a pleasant lighting apart from

saving the energy. High efficiency windows and insulation in walls involves ceilings and floors are used for the benefit of better temperature control.

A concept cross section of a green building is shown below

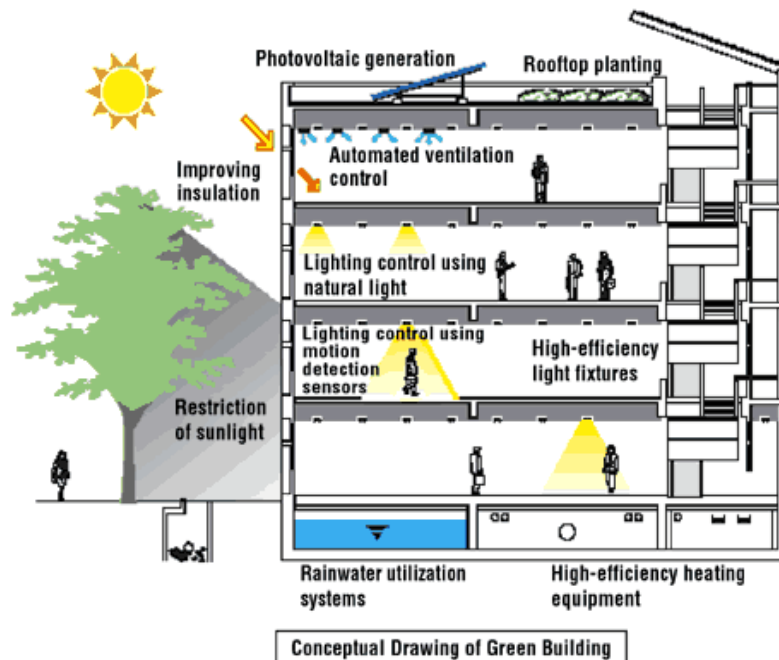


Fig. 4.2 Conceptual drawing of Green Building

2. Green Buildings improve internal environment for the occupants

- i. **Light:** In a designed green building the occupants shall feel as if they are in outdoor location. The interior and exterior designs shall integrate blending the natural and artificial lighting.
- ii. **Air:** in the air conditioned environment a green building shall be specially equipped to ensure the indoor air quality for a healthy atmosphere. Even the nasal feeling shall be pleasant free from the odour of paints and furnishings. This keeps people to be more productive.

3. Green Buildings preserve the environment at places far away from the buildings

Building materials like cement, sand, steel, stone, brick and a lot of finishing materials are responsible for about 20% of the greenhouse gases emitted by a building during its lifetime. Green Building shall use the products that are non-toxic, reusable, renewable and recyclable wherever possible. Locally manufactured products are preferred so that the collective material environment of the locality remains a constant. This also saves fuel for the transport of materials.

As in the case of food and domestic products are tagged with green as a fashion of eco-friendly practices; building materials are also going green. Here are some examples of futuristic green building materials which are in research stage now.

Green Wood: A Stanford team has found that hemp fibers and biodegradable plastic when pressed together and heated form layers and this material is as strong as wood. Also it degrades faster. This wood creates more raw materials when it breaks down. Microbes produce methane gas when they

decompose. This wood substitute and other debris thrown into landfills. Another type of bacteria absorbs this gas and turns it into plastic that can be used to create a new wooden plank. By this cycle, continuous source of raw material for this wood shall be ensured. Usage of this material may help to control deforestation.

Green cement: Bruce Constantz at Calera, based in Los Gatos, has developed a green method to produce both cement and aggregate. This method removes CO₂ from power plant flues and mixes the gas with seawater to produce the mineral raw materials of concrete. For every ton of green cement Calera manufactures half a ton of fly-ash from coal plants is used apart from preventing production and emission of CO₂.

Copper Slag: Researchers have suggested that slag from copper plants can be used as alternative for natural river sand. This reduces the mining of rivers and thereby lowering of water table of the locality.

Other Green Building materials : Renewable plant material like bamboo (because bamboo grows quickly) and straw, lumber from forest ecology blocks, dimension stone, recycled stone, recycled metal are some of the other materials used in a green building.

LEED green building rating system

Green buildings constructed should confirm to the prescribed standards. There should be continuous assessment and monitoring from the planning/design stage up to the completion of construction, for declaring a building as a green building. For this, LEED (Leadership in Energy and Environmental Design) green building rating system is allowed. In this system points are awarded for adapting green concepts in various categories and the buildings are certified green at levels such as silver, gold or platinum based on the total number of points they get in LEED rating. The details are shown below. (Table 4.5, 4.6)

Table 4.5: Audit for Green Concept in building:

Categories for rating	LEED points
Sustainable site	13
Water efficiency	6
Energy and atmosphere	17
Material and resources	13
Indoor Environment Quality	15
Design process and innovation	4
Employing LEED designer	1
Total points	69

Table 4.6 Certification for Green Buildings:

LEED Rating level	Points required
LEED Certified	26 to 32
Silver level	33 to 38
Gold level	39 to 51
Platinum level	52 and above

Green building- Benefits:

- Energy savings (20 to 30%)
- Water savings (30 to 50%)
- Enhanced air quality
- Excellent day lighting
- Health and well being of occupants
- Conservation of scarce resources.

ROADS- CLASSIFICATION OF RURAL AND URBAN ROADS

Introduction

Transportation is the process of transporting or carrying goods and passengers from one place to another. Transport is the medium used for transporting goods and passengers.

Transport system of a country is an indicator of its social, economical, industrial, commercial and cultural development. Good transport system helps in the movement of people and goods, enables contact between different places and improves the trade and industry considerably.

Modes of transport: The different types of transport available in our country are

1. Road ways
2. Rail ways
3. Air ways
4. Water ways
5. Pipelines

(1). Roadways: Road transport is the most important of all. Road transport is very flexible, cheap and easily available. Transportation of persons and light and heavy goods is possible with road transport. Road transport requires just moderate infrastructure facilities like bus stands, workshops and depots. Carts, buses, vans, etc are the vehicles that form the road transport. All other modes of transport depend on roadways.

(2). Rail ways: Rail transport is suitable and convenient for long distance movement of persons and goods. The capacity of rail transport is high. Even heavy goods can be transported by rail. Rail transport is more comfortable than other transports. Rail transport is reasonably fast. However, it requires heavy infrastructure facilities like tracks, platforms, shunting yards, passenger amenities, etc. In our country we have one of the largest networks of railway lines.

(3). Air ways: Air transport is through aircrafts. It is fastest of all. It is comfortable. Quick movement of persons, postal bags and important goods are possible with air transport. But it is not suitable for transporting very heavy goods. Air transport is very costly. Heavy infrastructure is required for air transport.

(4). Water ways: Water transport is the oldest form of transport. It is possible only where suitable watercourses like sea, calm rivers and streams and canals are available. Ships have very high carrying capacity and can transport extremely heavy goods. Water transport is cheap and moderately fast. Boats and canoes are used for local, short distance water transport like crossing of rivers, lakes and backwater etc.

(5) Pipe Lines: For transporting items like oil, gases, water etc. pipelines are used

Advantages of road transport

1. Road connect all the places in a country. This essentially facilitates the communication on land.

2. Roads accelerate the economical prosperity and general development of a country.
3. These serve as the feeder lines to other modes of transport. In other words, other modes of transport depend on road transport.
4. These help in administration and maintenance of law and order.
5. Road network are strategically important during wartime.
6. These help in growth of trade, industry, agriculture and phases of development.

Structure of the road

The road structure has the following components,

1. Sub grade (formation)
2. Sub base
3. Base (foundation)
4. Base coat
5. Wearing course

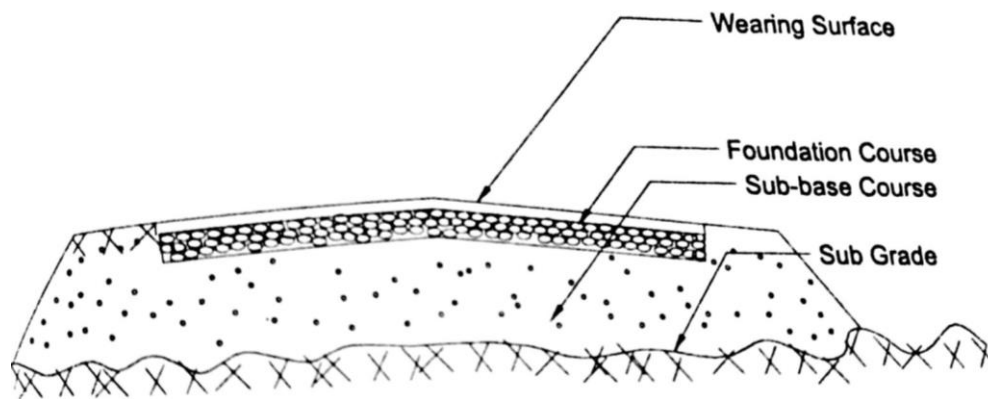


Figure4.3: Structure of the road

1. **Sub grade:** A well-compacted surface of earth on which the road pavement rests, is called *sub-grade or formation*. The sub-grades or road pavements may be either on natural ground, in cuttings or on embankments, according to the topography of the terrain and the required formation level.
2. **Sub-base:** A layer of cheap granular materials like burnt clinker, gravels are provided over the sub-grade, is called *sub-base*. It prevents the sub-grade soil to penetrate into base course. It checks capillary rise of sub-soil water and also improves the drainage and bearing capacity of the sub-grade.
3. **Base:** (not shown in figure): The layer of hard and durable stones provided over the base of a road pavement, is called *base coat*. Base coat may not be necessary in case of rocky sub-grade and base consisting of hard and durable stones.
4. **Wearing course:** The top most layer of the road pavement directly exposed to the vehicular traffic, is called *wearing course*. It provides smooth riding surface free from dust. It acts as an impervious layer to the road pavement and it distributes the traffic load to base course. It resists the abrasion caused due to the movement of vehicular traffic.

CLAASIFICATION OF ROADS:

Road can be classified on various aspects. The main classification is given here.

Classification based on location and function

Non-urban roads are classified based on their location, importance and functional responsibility. According to Nagpur plan of Indian Roads Congress (IRC), non-urban roads are classified as follows:

1. National Highways(NH)
 2. State Highways(SH)
 3. District Roads
 - a) Major District Roads(MDR)
 - b) Other District Roads(ODR)
 4. Village Roads(VR)
1. **National Highways:**

These are important road connecting state capitals, industrial canters, major-ports and foreign highways. NH roads run through the length and breadth of the country. A few NH roads are mentioned below:

- NH 1. Amritsar- Ambala- Delhi Road
- NH 2. Delhi- Agra- Kanpur- Calcutta Road
- NH 47. Kanyakumari- Kochi- Selam Road
- NH 17-Edappally-Mangalore Road

National Highways are constructed and maintained by C.P.W.D (Central Public Works Department) or M.E.S (Military Engineering Service) of our country. These roads are of at least two lanes and of width 8.0 m. A typical cross-section of a National Highway is shown in the Fig.4.5

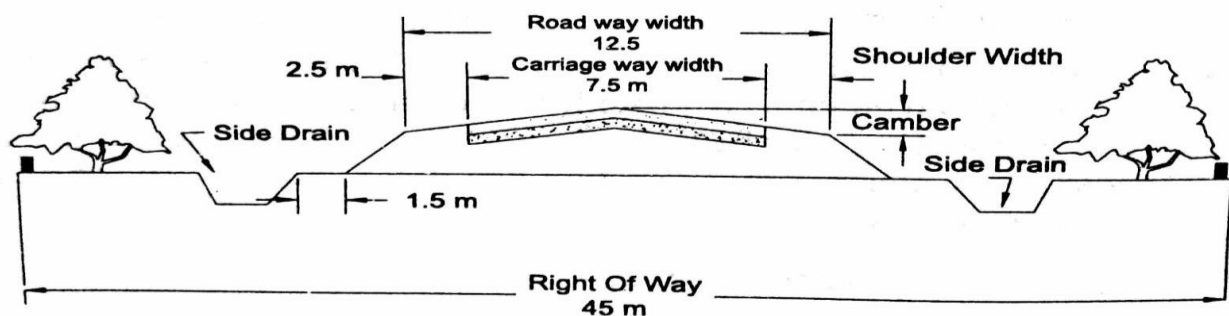


Fig 4.5 Cross section of a National Highway

2. State Highways:

These roads connect state capitals to the district head quarters and other major cities of a state. These are designed, constructed and maintained by State P.W.D. These are to be of the same standards of National Highways. Main Central Road (M.C) between Angamaly and Thiruvananthapuram is a state highway in Kerala.

3. District Roads:

These roads in a district connect the areas of production, markets each other or to National and State Highways. District roads will act as feeder roads from heart of rural areas to other important centres in the district. There are two types of district roads based on the order of traffic.

- a) Major District Roads(MDR)- for high order traffic
- b) Other District Roads(ODR)-for lower order traffic

4. Village Roads:

Village roads are connecting villages or groups of villages to the nearest district or state road. These are main channels of transport from village to arterial roads.

Classification of Urban roads

The classifications of urban roads according to Indian Roads Congress (IRC) are given below.

- 1. Express ways
- 2. Arterial streets
- 3. Sub- arterial streets
- 4. Collector streets
- 5. Local streets

1. Express ways

Express highways are major highways meant for vehicular traffic with full or partial access control (limited entry points) provided with grade separation at intersections. The main function is the movement of heavy volumes of traffic at high speeds under free flow conditions. Parking, loading and unloading of goods and pedestrian traffic are not permitted on express highways.

2. Arterial streets

Arterial streets are meant for through traffic (vehicles by pass city centers) usually on a continuous route. These roads serve as principle network for through traffic flow. These streets collect traffic from sub arterial and collector roads. Continuity is one of the important aspects of arterial road. Parking, loading and unloading of goods are restricted and pedestrians are allowed to cross only intersections.

3. Sub-Arterial streets

These are lower category of arterial streets in terms of traffic and importance. Parking, loading and unloading of goods are restricted and pedestrians are allowed to cross only at intersections. These are spaced about 0.5 km in central areas of city and 3 to 5 km at the outskirts.

4. Collector streets

Collector streets are meant for collecting and distributing traffic from local streets. They are located near residential colonies, business areas and industrial areas. There may be few parking restrictions during peak hours.

5. Local streets

These streets to access residential, business or other plots. Local streets do not carry heavy traffic. A local street allows unrestricted parking and pedestrian movements.

Classification based on materials and method of construction

- a) **Earth roads:** Earth roads are constructed with soil available at the site itself. Its foundation and wearing course are made with earth. It is cheap and used for light traffic. These roads are normally done at villages.
- b) **Soil stabilized roads:** The earth roads, which are stabilized by different methods, are called soil stabilized roads. Stabilization is done to improve the stability and durability of the road structure. The different methods of stabilization are mechanical stabilization, lime stabilization and bitumen stabilization.
- c) **Gravel roads:** One or two layers of gravel mixed with earth and sand are laid over a well compacted sub grade. The thickness of the layer may be 15 to 30 cm. These types of roads are called gravel roads.
- d) **Water bound Macadam roads:** Aggregates which are bonded with water and soil constitute this type of roads. Crushed stone, over burnt bricks, crushed slag, laterite or any other suitable aggregate are used in these roads. This road can be a base for bituminous roads, concrete roads. Both the base and wearing course of these roads are laid with aggregates of different sizes.
- e) **Bituminous roads:** When the binding material used in a road is bitumen, coal tar or asphalt, then it is called a bituminous road. These roads are impervious and dustless. Different types of bituminous roads are:
 - a) Surface dressed roads
 - b) Bituminous macadam
 - c) Sheet asphalt
 - d) Bituminous concrete road

In surface dressed roads, a thin layer of bitumen is applied over a water bound macadam or similar type. After that a thin layer of fine sand is spread over it.

Bituminous macadam roads are suitable for heavy traffic. After the base course is properly rolled, bitumen is allowed to penetrate into the layer of aggregates.

Sheet asphalt surfacing is used on the cement concrete base or water bound macadam. Over a layer of asphalt concrete, another layer of asphalt mortar is laid.

In bituminous concrete roads, coarse and fine aggregates are mixed with hot bitumen in mixer and laid over a suitable base. These roads are suitable for heavy traffic. These are good quality roads.

- f) **Concrete roads:** These are superior quality roads. Concrete roads are rigid pavements. It provide smooth riding surface under all conditions. These can be laid on a sub grade. It has

low maintenance cost and better weather resisting qualities. There are two types of concrete roads.

1. Premixed concrete roads
2. Cement bound macadam roads

Premixed concrete roads are constructed as slab construction. Concrete is mixed and laid on the prepared bed of sub grade.

In cement bound macadam roads, aggregates are laid and rolled to the desired thickness. Then cement slurry is spread over it.

Cross section of roads

A typical cross section of road is shown in the figure.4.6. The important elements of the road are

1. Right of way
2. Road way width
3. Carriage way
4. Shoulder width
5. Side drain
6. Camber

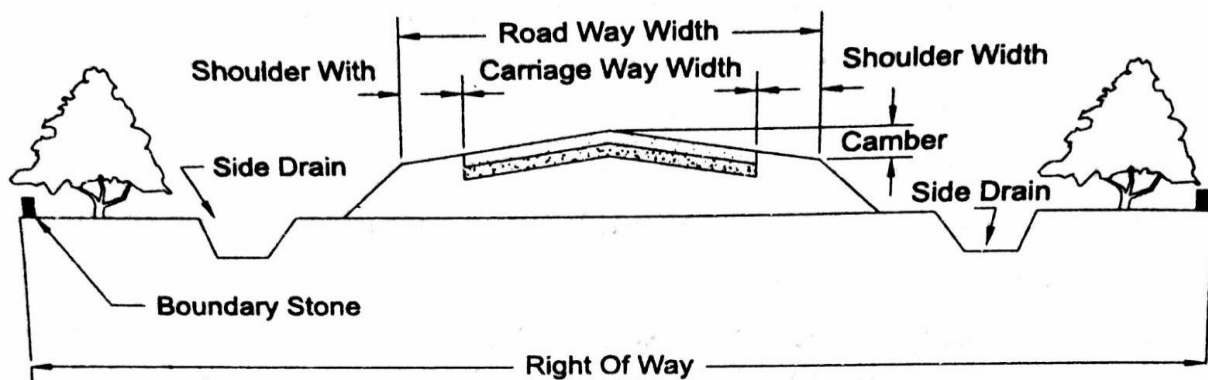


Fig 4.6 Cross sectional elements of road(NH)

- a) **Right of way:** The width of the land acquired for road purposes for the future developments is called right of way. This depends on the type of the road, terrain and other topological features. This land is solemnly under the highway authorities.
- b) **Roadway width:** The total width of formation is called roadway width. It includes the width of the carriage way and width of shoulders on both sides. This is also called formation width.
- c) **Carriage way:** The surfaced width used for vehicular traffic is known as carriage way. Indian Roads Congress stipulates a minimum width of 3.75m for single lane and 7.5m for double lane. A width of 3.5m is to be added for every additional lane.
- d) **Shoulder width:** The width provided between outer edge formation and carriageway is

called shoulder width. Normally a width of 2.5m is adopted for highways.

- e) **Side drain:** Side drain is provided at both sides of the road to facilitate the longitudinal drainage.
- f) **Camber:** Camber is the rise provided at the centre of the road to facilitate the lateral drainage. It is also called as cross fall or transverse slope. Camber to be provided depends on
 - 1. Nature of the surface of carriageway
 - 2. The intensity of rainfall of the region
- g) **Super elevation:** Super elevation is the raising of outer edge of a road than the inner edge, at the horizontal curves. It is also called as banking or cant. When a vehicle negotiates a horizontal curve, centrifugal forces are developed. This will tend the vehicle to topple over at curves. The horizontal component the weight balances the centrifugal force. This can be calculated using a formula

$$\text{Super elevation, } e = V^2 / 127R$$

Where, V – Design speed in Kmph

R – Radius of the curve in meters

SOURCES OF WATER- WATER SUPPLY –QUALITY OF WATER

TYPES OF SOURCES FOR WATER SUPPLY SCHEMES

The sources from which water is available for water supply schemes can conveniently be classified into the following two categories according to their proximity to the ground surface.

1. Surface sources
2. Underground sources

We will now discuss various forms of surface sources and underground sources.

1. SURFACE SOURCES FOR WATER SUPPLY SCHEMES

In this type of source, the surface runoff is available for water supply schemes. The usual forms of surface sources are as follows

- | | |
|----------------------|-----------------------|
| 1. Lakes and streams | 2. Ponds |
| 3. Rivers | 4. Storage reservoirs |

Each of the forms of surface sources will now be briefly discussed.

1. Lakes and streams: A natural lake represents a large body of water within land with impervious bed. Hence, it may be used as a source of water supply scheme for nearby localities. The quantity of runoff that goes to the lake should be accurately determined and it should be seen that it is at least equal to the expected demand of locality. Similar is the case with streams which are formed by the surface runoff.

It is found that the flow of water in streams is quite ample in rainy seasons but it becomes less and less in hot seasons and sometimes the stream may even become absolutely dry.

2. Ponds: A pond is a man-made body of standing water smaller than a lake. Thus the ponds are formed due to excessive digging of ground for the construction of roads, houses etc and they are filled up with water in the rainy seasons. The quantity of water in ponds is very small and it contains many impurities.

A pond cannot be adopted as a source of water supply and its water can only be used for washing of clothes or for animals only.

3. Rivers: The large rivers constitute the principal source of water supply schemes of many cities. Some rivers are perennial while others are non- perennial. The former rivers are snow fed and hence, the water flows in such river in all the seasons. The latter type of rivers dries in summer either wholly or partly and in monsoons, the heavy floods visits them. For such type of rivers, it is desirable to store the excess water of floods in monsoons by constructing dams across such rivers. The stored water may then be used in summer.

The principal use of a river can be summarised as follows

- i. It can be developed as the chief source of water supply for a town or a city.
- ii. It can be used for navigation.
- iii. It can be used to supply water for irrigation purposes.
- iv. It can serve as an agent of purification of waste.

- v. It can serve as centre of recreational activities such as bathing, boating, fishing, fountains etc

4. Storage reservoirs: An artificial lake formed by the construction of dam across a valley is termed as a storage reservoir. Whatever may be the size or use of a reservoir, the main object or function of a reservoir is to store water and thus it stabilises the flow of water.

The most important physical characteristic of a reservoir is therefore its storage capacity. The topographic survey of the dam site is carried out and a contour map is prepared. The capacity of reservoir is then worked out with the help of a contour map.

A storage reservoir(Fig 4.7) essentially consists of the following 3 parts

- i. A dam to hold water
- ii. A spillway to allow the excess water to flow; and
- iii. A gate chamber containing necessary valves for regulating the flow of water.

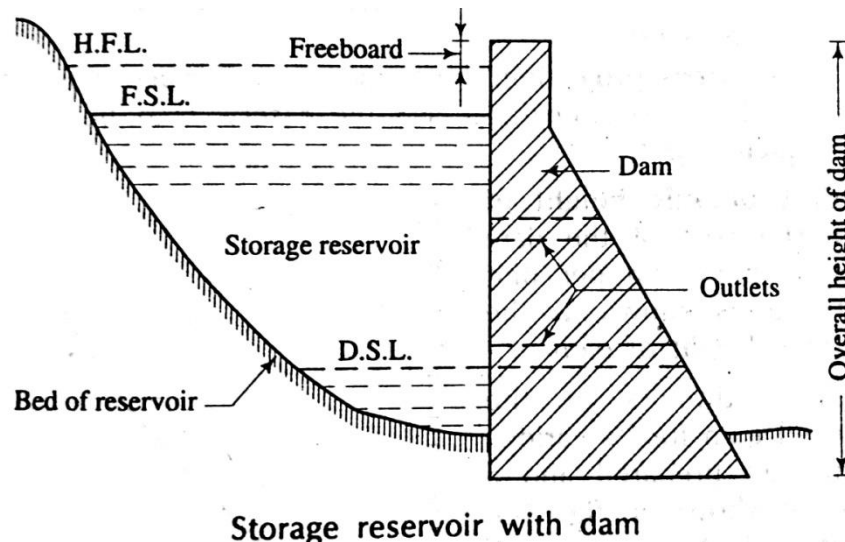


Fig 4.7 Storage Reservoir

2. UNDERGROUND SOURCES FOR WATER SUPPLY SCHEMES

In this type of source, the water that has percolated into the ground is brought on the surface. The difference between the terms infiltration and percolation should be noted. The entrance of rain water or melted snow into the ground is referred as the infiltration. The movement of water after entrance is called the percolation.

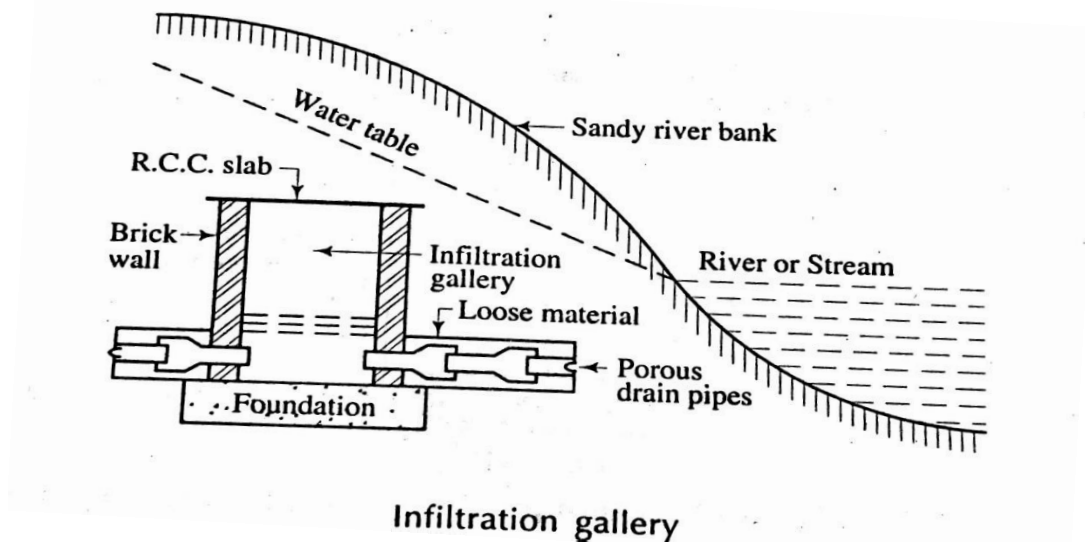
Forms of underground sources are

1. Infiltration galleries
2. Infiltration wells
3. Springs
4. Wells

Each of the above form of the ground sources will now be separately discussed in brief.

1. Infiltration galleries: An infiltration gallery is a horizontal or nearly horizontal tunnel which is constructed through water bearing strata. It is sometimes referred to as the horizontal well(Fig 4.8).

The gallery is usually constructed of brick walls with slab roof as shown in the figure. The gallery obtains its water from the water bearing strata by various porous drain pipes. These pipes are covered with gravels, pebbles etc. so as to prevent the entry of very fine material into the pipe.



Fig

4.8

The gallery is laid at a slope and the water collected in the gallery is led to a sump from where it is pumped and supplied to consumers after proper treatment. The manholes are provided along the infiltration gallery for the purpose of cleaning and inspection.

The infiltration galleries are useful as source of water supply when groundwater is available in sufficient quantity just below ground level or so. The galleries are usually constructed at depth of 5-10 m from the ground level.

2. Infiltration wells: In order to obtain large quantities of water, the infiltration wells are sunk in series in the banks of rivers. The wells are closed at top and open at bottom. They are constructed of brick masonry with open joints as shown in the figure.4.9

For the purpose of inspection of well, the manholes are provided in the top cover. The water infiltrates through the bottom of such wells and as it has to pass through sand bed, it gets purified to some extent.

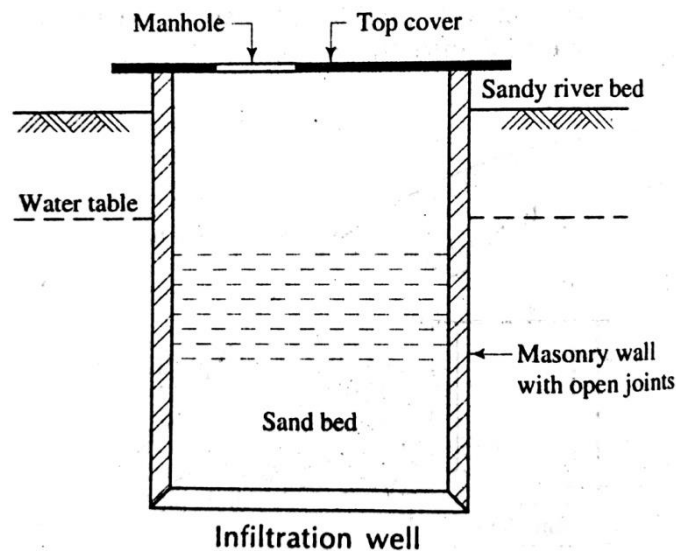


Fig 4.9. Infiltration well

3. Springs: When groundwater appears at the surface for any reasons, the springs are formed. They serve as source of water supply for small towns, especially near hills or bases of hills. Some springs discharge hot water due to the presence of sulphur and other minerals in their formations. These hot springs cannot be used to supply water for domestic purposes. But the hot water is found to cure some of the human disorders.

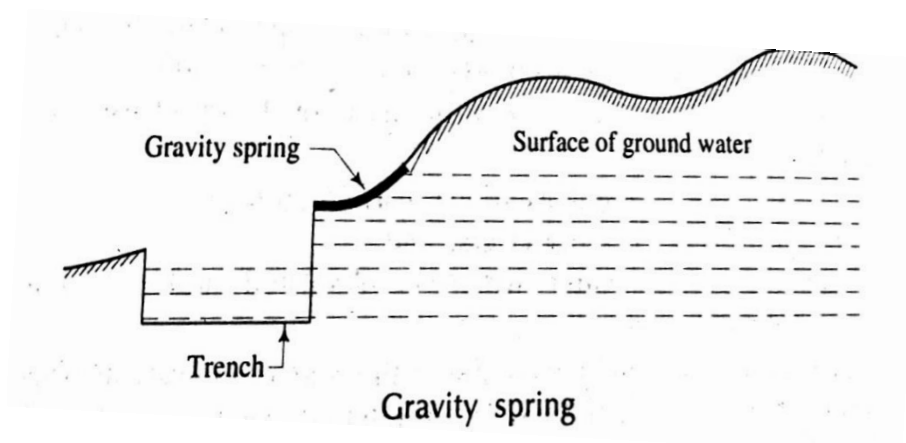


Fig 4.10. Gravity Spring

4. Wells: A well is defined as an artificial hole or pit made in the ground for the purpose of tapping water. The holes made for tapping oil are also known as the wells but in the general sense, a well indicate a source of water. In India, the cheap source of water supply for most of its population is wells and it is estimated that 75-85% of Indian population has to depend on wells for its water supply.

The 3 factors which form the basis of theory of wells are as follows:

- i. Geological conditions of the earth's surface
- ii. Porosity of various layers, and
- iii. Quantity of water which is absorbed and stored in different layers.

WATER SUPPLY SYSTEMS

The primary objective of water treatment for public supply is to take water from the best available source and to subject it to processing which will ensure water of good physical quality, free from unpleasant taste or odour and containing nothing which might be detrimental to health. The treatment of water to improve its quality involves additions to, subtractions from, or chemical changes in the raw water. Municipal water systems consist of the following units.

1. Collection works.
2. Transmission works
3. Purification works
4. Distribution works

These systems are shown diagrammatically below (Fig 4.11)

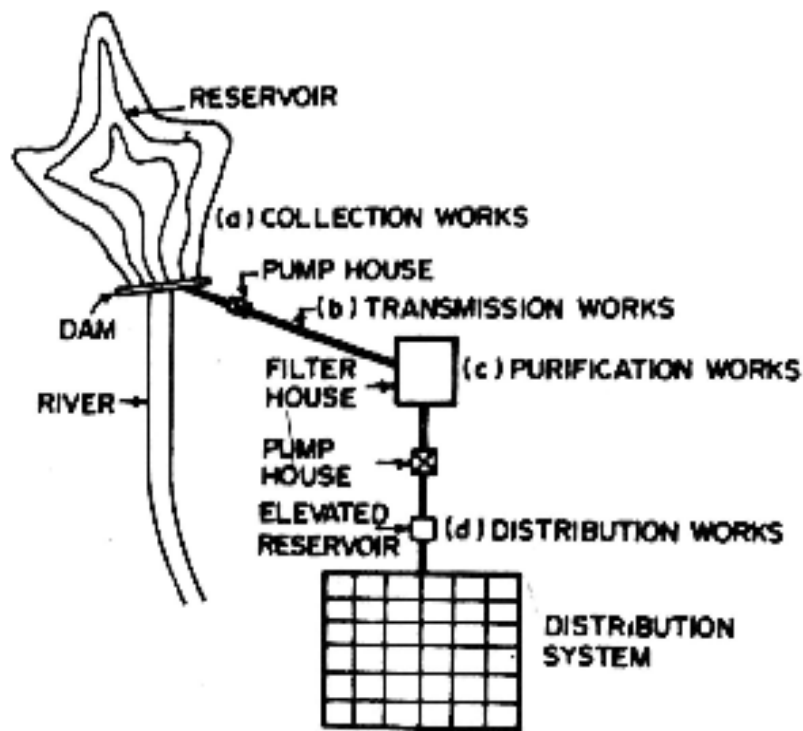


Fig .4.11 Water supply system

1. Collection works

Collection works are meant for the development of surface water or ground water resources. For major cities, or where water requirements are large, water is collected from a surface source - mostly a river or stream. If the river is perennial, a direct intake structure can be built on the river bank. If, however, river is not perennial, a dam is built across the river so that water is stored in the reservoirs. Water is then drawn from the reservoir as per needs. The collection works, therefore, consists of a storage or diversion work, and an intake structure.

2. Transmission works

In many cases, the collection works may be far away from the city where water is to be

supplied. In that case, water is conveyed to the city through the transmission works. These form the connecting link between the collection works and the purification works. Depending upon the topography of the area between the two sites, the transmission works may be in the form of conduits, canals or aqueducts. For simply gravity flow, canals are generally used. However if highlands intervene, pumping may have to be resorted to.

3. Purification works.

The water collected directly from the source may not be safe for drinking because of physical, chemical and biological impurities. The municipal water works must deliver to the consumer the water that is:

- i. Hygienically safe
- ii. Aesthetically attractive and palatable,
- iii. Economically satisfactory for its intended use.

Diseases like typhoid, cholera, dysentery etc, are water borne diseases. The principal aim of the purification works is to supply clean and bacteria free water. The common components of water purification works are

- i. Filtration plants to remove objectionable colour, turbidity, bacteria and other harmful organisms,
- ii. Deferrization and demanganization plants to remove excessive amounts of iron and manganese, and
- iii. Softening plants to remove excessive amounts of scale forming, soap consuming ingredients like calcium and magnesium ions.

4. Distribution works.

The treated and purified water is finally sent to the consumers through suitable distribution system. In order that water may flow in the water supply pipes under pressure, the purified water is normally stored in an elevated service reservoir. More than one reservoir may be needed in large systems. There are two patterns of water distribution system:

- (i) Branching pattern with dead ends, and
- (ii) Grid iron pattern.

The plan, topography and location of the area with respect to the service reservoir establish the type of distribution system and its character of flow.

QUALITY OF WATER

Pure water is not found in nature. It may contain certain impurities like sand, silt, clay and other inorganic matter. The impurities in water may be classified as follows:

- 1. Suspended impurities:** The solid particles, which are dispersed in water and large enough to be removed by filtration or settling by sedimentation are known as suspended impurities. These impurities are microscopic and include algae, protozoa, fungi (which causes colour, odour and turbidity). Silt, clay, mineral matter, organic and inorganic matter (which causes turbidity) and bacteria (which causes disease). The suspended impurities cause turbidity in water. The concentration of suspended matter in water is measured by its turbidity.
- 2. Colloidal impurities:** The finely divided dispersions of solid particles which are not visible in the naked eye and cannot be removed by ordinary filters are known as colloidal impurities. These impurities if associated with organic matter having bacteria become the chief source of epidemic. The colour of water is mostly due to the presence of colloidal impurities. The amount of colloidal impurities is determined by colour tests.
- 3. Dissolved impurities:** the impurities like organic compounds, inorganic salts and gases in the form of solids, liquids and gases which are dissolved in water are known as dissolved impurities. The following are the various dissolved impurities and their effects.
 - a) Salts
 - i. Calcium and magnesium chlorides, carbonates and bicarbonates cause hardness and alkalinity in water.
 - ii. Sodium chloride, carbonate and bicarbonate cause softening of water.
 - b) Metals
 - i. Iron causes red colour in water, increases hardness and corrosion.
 - ii. Manganese causes brown colour
 - iii. Lead and arsenic causes poisoning
 - c) Gases
 - i. Oxygen corrodes the metal
 - ii. Chlorine and ammonia causes bad taste and odour.
 - iii. Carbon dioxide and hydrogen sulphide causes acidity in water.

Indian standard specification for drinking water (as per IS 10500) is shown in the table 4.7

Table.4.7

**INDIAN STANDARD SPECIFICATIONS FOR DRINKING WATER
IS: 10500**

S.NO.	Parameter	Requirement desirable Limit	Remarks
✓ 1.	Colour	5	May be extended up to 50 if toxic substances are suspected
✓ 2.	Turbidity	10	May be relaxed up to 25 in the absence of alternate
✓ 3.	pH	6.5 to 8.5	May be relaxed up to 9.2 in the absence
✓ 4.	Total Hardness	300	May be extended up to 600
5.	Calcium as Ca	75	May be extended up to 200
6.	Magnesium as Mg	30	May be extended up to 100
✓ 7.	Copper as Cu	0.05	May be relaxed up to 1.5
8.	Iron	0.3	May be extended up to 1
9.	Manganese	0.1	May be extended up to 0.5
✓ 10.	Chlorides	250	May be extended up to 1000
11.	Sulphates	150	May be extended up to 400
12.	Nitrates	45	No relaxation
✓ 13.	Fluoride	0.6 to 1.2	If the limit is below 0.6 water should be rejected. Max. Limit is extended to 1.5
14.	Phenols	0.001	May be relaxed up to 0.002
15.	Mercury	0.001	No relaxation
16.	Cadmium	0.01	No relaxation
17.	Selenium	0.01	No relaxation
✓ 18.	Arsenic	0.05	No relaxation
19.	Cyanide	0.05	No relaxation
✓ 20.	Lead	0.1	No relaxation
21.	Zinc	5.0	May be extended up to 10.0
22.	Anionic detergents (MBAS)	0.2	May be relaxed up to 1
23.	Chromium as Cr ⁺⁶	0.05	No relaxation
24.	Poly nuclear aromatic Hydrocarbons	--	--
25.	Mineral Oil	0.01	May be relaxed up to 0.03
26.	Residual free Chlorine	0.2	Applicable only when water is chlorinated
✓ 27.	Pesticides	Absent	--
28.	Radio active	--	--

